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SEAT No. :

P6988

[5865] - 301

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S.Y. M.C.A. (Management)

IT - 31: MOBILE APPLICATION DEVELOPMENT

(2020 Pattern) (Semester - III)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.*
- 2) Figures to the right indicate full marks.*

Q1) Multiple Choice Questions:

[10]

For each of the following multiple choice questions, choose the correct answer.

- i) Android is based on which kernel.
 - a) Linux
 - b) Windows
 - c) Mac
 - d) Redhat
- ii) What is an Activity?
 - a) A single screen with UI
 - b) A layout
 - c) A Fragment
 - d) None of the above
- iii) In which Directory XML Layout Files are stored?
 - a) /assets
 - b) /src
 - c) /res/values
 - d) /res/layout
- iv) ADB stands for
 - a) Android Driver Bridge
 - b) Android Debug Bridge
 - c) Android Delete Bridge
 - d) Android Destroy Bridge
- v) An Android _____ is an abstract description of an operation to be performed. It can be used to start Activity to launch an activity.
 - a) Filters
 - b) Intent
 - c) Service
 - d) Broadcast Receiver
- vi) SQLite is a?
 - a) No SQL database
 - b) Distributed database
 - c) Relational database
 - d) Operational database

P.T.O.

- vii) Which of the following is not a layer in Android Architecture?
- a) Libraries
 - b) Linux kernel
 - c) Android Runtime
 - d) Application Framework
- viii) The extras can be set and read using the?
- a) setExtras () and getExtras () methods
 - b) putExtras () and getExtras () methods
 - c) setsExtras () and getsExtras () methods
 - d) None of the above
- ix) Which of the following is Application development environments for iOS?
- a) Cocoa
 - b) Cocoa touch
 - c) Cocoa iOS
 - d) Cocoa begin
- x) _____ sets the gravity of a view or layout in its parent.
- a) Android : gravity
 - b) Android : Layout-gravity
 - c) Android : weight
 - d) Android : orientation
- xi) Among the following choose the one which convert Java byte code into Dalvik byte code.
- a) Dex compiler
 - b) Mobile Interpretive compiler
 - c) Dalvik converter
 - d) None of the above
- xii) _____ means it will occupy the complete space available on the display of the device.
- a) Wrap-content
 - b) Match-content
 - c) Match-parent
 - d) Wrap-parent
- xiii) React Native ApI accepts two methods as
- a) alert (), prompt ()
 - b) get (), post ()
 - c) get (), alert ()
 - d) post (), prompt ()
- xiv) What does ES6 stands for?
- a) ECMA Script 6
 - b) ECMA 6
 - c) ECMA Javascript 6
 - d) EJavascript 6

- Q3)** a) What is Activity? Explain Activity Life cycle. [6]
b) Demonstrate Array Adapter using Listview to display list of five subjects. [6]

OR

- a) What is Fragment? Explain Fragment Life cycle. [6]
b) Design android application for login activity. Write an android code to check login credentials with username = “SPPU” and password = “Exam”. Display appropriate message on Toast to the user. [6]

- Q4)** a) Write an android application using SQLite to create employee (eid, ename, edept, esalary) table and insert a record in table and display appropriate message on toast to user. [6]
b) Write a short note on JSON parsing with example. [6]

OR

- a) Explain cursor in android with suitable example. [6]
b) Demonstrate webview to display a webpage in an android application. [6]

- Q5)** a) What is Flutter? Explain Architecture of Flutter. [6]
b) Explain Android Menu. [4]

OR

- a) What is React Native? State the advantages of React Native. [6]
b) What is view? Explain any two views with suitable Example. [4]



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M.C.A. (Management)

IT 32: DATA WAREHOUSING AND DATA MINING

(2020 Pattern) (Semester - III)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.*
- 2) Draw neat & labelled diagram wherever necessary.*

Q1) Answer the following multiple choice questions. **[20×½=10]**

- i) Find the wrong statement of the K-means clustering.
 - a) K-means clustering is a method of vector quantization.
 - b) K-means neighbour is same as K-nearest.
 - c) K-means clustering aims to partition 'n' observations into K clusters.
 - d) K-means clustering produces the final estimate of cluster centroids.

- ii) How many tier data warehouse architecture?
 - a) 2
 - b) 1
 - c) 3
 - d) 4

- iii) _____ is an intermediate storage area used for data processing in ETL process of data warehousing.
 - a) Buffer
 - b) Virtual memory
 - c) Staging area
 - d) Inter storage area

- iv) _____ is a good alternative to the star schema.
 - a) Star schema
 - b) Snowflake schema
 - c) Fact constellation
 - d) Star-snowflake schema

- v) In a snowflake schema which of the following types of tables are considered?
 - a) Fact
 - b) Dimension
 - c) Both fact and dimension
 - d) None of the mentioned

P.T.O.

- vi) The role of ETL is to _____.
 a) Find erroneous data
 b) Fix erroneous data
 c) Both finding and fixing erroneous data
 d) Filtering of the data source
- vii) _____ is a data transformation process.
 a) Comparison b) Projection
 c) Selection d) Filtering
- viii) OLTP stands for _____.
 a) Online Transaction Protocol
 b) Online Transaction Processing
 c) Online Terminal Protocol
 d) Online Terminal Processing
- ix) An approach in which the aggregated totals are stored in a multidimensional database while the detailed data is stored in the relational database is a _____.
 a) MOLAP b) ROLAP
 c) HOLAP d) OLAP
- x) Summary of data from an OLAP can be presented in _____.
 a) Normalization b) Primary keys
 c) Pivot Table d) Foreign keys
- xi) Efficiency and scalability of data mining algorithm is related to _____.
 a) Mining methodology b) User interaction
 c) Diverse data types d) None of the mentioned
- xii) Strategic value of data mining is _____.
 a) Cost sensitive b) Work sensitive
 c) Time-sensitive d) Technical-sensitive
- xiii) If the ETL process fetches the data separately from the host server during the automatic load to the data warehouse, one of the challenge involved is _____.
 a) the associated network may be down
 b) it may end up pulling the incomplete /incorrect file.
 c) it may end up connecting to an incorrect host server.
 d) None

- xiv) Webmining helps to improve the power of web search engine by identifying _____
- a) Web pages and classifying the web documents
 - b) XML documents
 - c) Text documents
 - d) Database
- xv) _____ is achieved by splitting the text into white spaces.
- a) Text cleanup
 - b) Tokenization
 - c) Speech tagging
 - d) Text transformation
- xvi) Assigning data to one of the clusters and recompute the centroid are the steps in which algorithm.
- a) Apriori algorithm
 - b) Bayesian classification
 - c) FP tree algorithm
 - d) K-mean
- xvii) A disadvantage of KNN algorithm is, it takes _____
- a) More time for training
 - b) More time for testing
 - c) Equal time for training
 - d) Equal time for testing
- xviii) Which is not a characteristics of Data warehouse?
- a) Volatile
 - b) Subject oriented
 - c) Non volatile
 - d) Time variant
- xix) What is the first stage of Kimball Life Cycle diagram.
- a) Requirement Definition
 - b) Dimensional Modelling
 - c) ETL Design Development
 - d) Maintenance
- xx) A connected region of a multidimensional space with a comparatively high density objects.
- a) Clustering
 - b) Association
 - c) Classification
 - d) Subset

Q2) a) What is a Data warehouse. Explain the need and characteristics of Data warehouse. [5]

b) Explain the schemas of Data warehouse. [5]

OR

a) Explain Kimball Life Cycle diagram in detail. [5]

b) What is a Data warehouse? Explain the properties of Data warehouse architecture. [5]

Q3) a) What is ETL? Explain data preprocessing techniques in detail. [6]

b) What is OLAP? Describe the characteristics of OLAP. [4]

OR

a) Describe ETL. What are the tasks to be performed during data transformation. [6]

b) What are the basic operations of OLAP? [4]

Q4) a) What is Data mining? Explain the architecture of Data mining. [3]

b) Apply FP Tree Algorithm to construct FP Tree and find frequent itemset for the following dataset given below (minimum support = 30%) [7]

Transaction ID	List of Products
1	Apple, Berries, Coconut
2	Berries, Coconut, Dates
3	Coconut, Dates
4	Berries, Dates
5	Apple, Coconut
6	Apple, Coconut, Dates

OR

- a) Explain data mining techniques in brief. [3]
- b) How does the KNN algorithm works? [7]

Apply KNN classification algorithm for the given dataset and predict the class for $X(P_1 = 3, P_2 = 7)$ ($K = 3$)

P_1	P_2	Class
7	7	False
7	4	False
3	4	True
1	4	True

- Q5)** a) What is text mining? Explain the process of text mining. [4]
- b) Explain K-means algorithm. Apply K-means algorithm for group of visitors to a website into two groups using their age as follows:
 15, 16, 19, 20, 21, 28, 35, 40, 42, 44, 60, 65
 (Consider initial centroid 16 and 28 of two groups) [6]

OR

- a) Apply K-means algorithm for the given data set where K is the cluster number $D = \{2, 3, 4, 10, 11, 12, 20, 25, 30\}$, $K = 2$. [6]
- b) What are the different types of web mining? [4]



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S.Y. M.C.A. (Management)

**IT 33 : SOFTWARE TESTING AND QUALITY ASSURANCE
(2020 Pattern) (Semester - III)**

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates: .

- 1) Q.1 & Q.7 are compulsory.*
- 2) Solve any four questions from Q.2 to Q.6.*

Q1) a) Write detail test plan for online registration & examination application. Candidate will register through valid email ID & will book date & time slot from available dates & time slots. Candidate will select subject from list of subjects given. Application will send user ID & password to the candidates email ID. On the date & time, candidate will login with given user ID & Password Application will display multiple choice questions & will accept choice from candidate. At the end of given time, application will display the marks. **[8]**

b) Design suitable test cases for above application. **[4]**

Q2) Define software quality assurance. Also explain SQA activities. **[8]**

Q3) Explain various techniques of software reviews. Also compare them. **[8]**

Q4) Explain various black box testing techniques with reference to entry of MCA sem III marks data entry. **[8]**

Q5) Explain type of testing tools. Also explain their benefits & risks. **[8]**

Q6) Explain non functional testing. Also compare localization & internationalization. **[8]**

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Q7) Write short note on any two.

[6]

- a) Software quality metrics.
- b) 'V' model of testing.
- c) Experience based testing.
- d) Defect life cycle.



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P6992

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[Total No. of Pages : 4

S.Y. M.C.A. (Management Faculty)
IT - 35: CLOUD COMPUTING
(2020 Pattern) (Semester - III)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.*
- 2) Figures to the right indicate full marks.*

Q1) Write correct option or filling the blanks of following questions.(each ½mark)

[10]

- i) Which one of the following is cloud platform by microsoft?
 - a) Azure
 - b) AWS
 - c) Cloudera
 - d) None of the above
- ii) Which of the following are the features of cloud computing?
 - a) Security
 - b) Availability
 - c) Large Network access
 - d) All of the mentioned
- iii) Which of the following is a type of cloud computing services?
 - a) Service-as-a-Software (SaaS)
 - b) Software-and-a-Server (SaaS)
 - c) Software-as-a-Service (SaaS)
 - d) Software-as-a-Server (SaaS)
- iv) Who is the father of cloud computing?
 - a) Sharon B. Codd
 - b) Edgar Frank Codd
 - c) J.C.R. Licklider
 - d) Charless Bachman

P.T.O.

- v) Which of the following is not a type of cloud server?
 - a) public cloud servers b) private cloud servers
 - c) dedicated cloud servers d) merged cloud servers

- vi) Which one of the following cloud concepts is related to sharing and pooling the resources?
 - a) Polymorphism b) Virtualization
 - c) Abstraction d) None of the mentioned

- vii) Which of the following is an essential concept related to cloud?
 - a) Reliability b) Abstraction
 - c) Productivity d) All of the mentioned

- viii) Which of the following type of virtualization is also characteristic of cloud computing?
 - a) Storage b) Application
 - c) CPU d) All of the mentioned

- ix) Which of the following is used to aid for locating services in SOA?
 - a) databus b) Catalog service
 - c) Data abstraction services d) All of the mentioned

- x) How many different protocol types are required for message passing in SOA?
 - a) 2 b) 4
 - c) 1 d) 3

- xi) Which is the Need of SOA?
 - a) Scalability b) Adaptivity
 - c) Robustness d) None of these

- xii) A _____ level Agreement (SLA) is the contract for performance negotiated between you and a service provider.
 - a) Application b) Service
 - c) Deployment d) All of the mentioned

- xiii) Combination of cloud computing architecture is _____.
a) service-oriented architecture and grid computing
b) utility computing and event-driven architecture
c) virtualization and event-driven architecture
d) service-oriented architecture and event-driven architecture
- xiv) Broadly divide the cloud architecture into?
a) 2
b) 3
c) 4
d) 5
- xv) The data used by multiple processor is called _____.
a) Shared data
b) Protected data
c) Private data
d) Public data
- xvi) Which of these is Identity in IAM?
a) Users
b) Groups
c) Roles
d) All of these
- xvii) IAM group:
a) Is same as IAM users
b) Can be used to specify permission for a collection of users
c) Is truly an identity
d) All of these
- xviii) Which of the following service provider provides the least amount of built in security?
a) SaaS
b) PaaS
c) IaaS
d) All of the mentioned
- xix) What are the features of docker?
a) Easy modelling
b) Version control
c) Placement
d) All of the above
- xx) Identity the component of the docker?
a) Image
b) Platform image
c) Container
d) All of the above

Q2) What are benefit and challenges of cloud computing? Explain the term of SaaS, PaaS & IaaS. [10]

OR

What is cloud computing? Explain in details cluster computing and Grid computing? [10]

Q3) a) What is the Virtualization & write the type of Virtualization? [5]

b) Define the Service Oriented Architecture and the concept of SOA. [5]

OR

a) What is Virtual Machine (VM)? Write on open source virtualization manager? [5]

b) Write short note on cloud Governances. [5]

Q4) a) Write short note on the Cloud Computing Architecture? [5]

b) Write the data migration in cloud strategies & process? [5]

OR

a) Write the multi-cloud Architecture and its benefit & challenges? [5]

b) What is the concept of cloud Bursting? [5]

Q5) Explain the cloud computing privacy and security challenges in details? [10]

OR

Explain in details the identity management & Access control? [10]

